

Titan Core Engine Stack (Canonical)

Naming Rule

All core engines follow:

Titan + Function + Engine

This keeps the platform readable, consistent, and easy to onboard developers and AI agents into.

1. Titan Identity Engine

The security and tenancy spine of the platform.

Responsible for:

users

employees

teams / companies

roles & permissions

authentication

device identity

tenant boundaries

Typical tables (platform, no prefix):

users

teams

team_users

roles

permissions

user_devices

Rule:

Everything else must resolve identity + tenancy through this engine.

2. Titan Customer Engine

The customer and property management engine.

Responsible for:

customers

service locations / sites

addresses

preferences

service history

access notes / hazards / keys

Core tables (tz_):

tz_customers

tz_customer_events

tz_sites

tz_site_access

tz_site_keys

tz_site_hazards

tz_site_rooms

This engine connects the real world to work execution.

3. Titan Work Engine

The operational execution engine.

Responsible for lifecycle of work performed for customers:

jobs

visits

assignments

tasks

checklists

execution states

work events

Core tables (tz_):

tz_jobs

tz_job_visits

tz_job_events

tz_job_assignments

tz_tasks

tz_task_events

tz_job_checklists

tz_job_checklist_items

tz_job_checklist_responses

This engine answers:

What work exists?

Who is doing it?

What state is it in?

4. Titan Evidence Engine

The proof and compliance engine.

Responsible for recording and linking:

photos

videos

signatures

incident reports

compliance evidence

proof bundles

Core tables (tz_):

tz_evidence_chain

tz_evidence_items

tz_evidence_links

This engine makes the system legally defensible and auditable.

5. Titan Docs Engine

The document and artifact engine.

Handles:

business documents

quotes and reports (as documents)

evidence bundles

contracts

document versioning and linking

Core tables (tz_):

tz_documents

tz_document_versions

tz_document_links

Your evidence vault and document gallery sit here.

6. Titan Money Engine

The financial lifecycle engine.

Responsible for:

quotes

invoices

payments

refunds

financial events

overdue collections

Core tables (tz_):

tz_quotes

tz_quote_items

tz_invoices

tz_invoice_items

tz_payments

tz_payment_events

tz_financial_events

This engine answers:

What money is owed?

What money has been received?

7. Titan Talk Engine

The communications engine.

Handles:

customer messaging

staff messaging

omnichannel inbox

message attachments

thread linking to work and money

Core tables (tz_):

tz_threads

tz_messages

tz_message_attachments

tz_thread_links

tz_contacts

Channels like email/SMS/WhatsApp/Telegram/webchat plug into this engine.

8. Titan Signal Engine

The event and signal bus of the platform.

Rule:

Every meaningful system action emits a signal.

Signals are append-only.

Examples:

work.job.created

work.job.completed

finance.invoice.overdue

customer.message.received

evidence.uploaded

Core tables (tz_):

tz_signals

tz_signal_events

tz_signal_consumers

Consumers include:

Titan Pulse Engine

Titan Zero Engine

Analytics listeners

Extension listeners

Clarification:

Titan Signal records what happened. It does not decide what to do.

9. Titan Pulse Engine

The automation and orchestration engine.

Responsible for:

running rules

executing workflows

chaining actions across engines

Examples:

job completed → generate invoice

invoice overdue → send follow-up

job tomorrow → notify worker

complaint → create rework job

Core tables (tz_):

tz_automation_rules

tz_automation_triggers

tz_automation_runs

tz_automation_logs

Clarification:

Titan Pulse executes actions and writes to operational tables only through engine APIs (Work, Money, Talk, Docs, Evidence). Pulse should not directly mutate random tables outside engine boundaries.

10. Titan Zero Engine

The AI reasoning and governance engine.

Responsible for:

AI assistants

reasoning + planning

artifact generation

model routing (multi-modal open router)

approvals and decision oversight

Internal cognitive architecture:

Titan Zero Core (central orchestration)

OmegaCore (planning + strategic big picture)

OmicronCore (fine detail validation)

Modes:

Default: Titan Zero Mode (monitoring + routine guidance)

Risk-triggered: Trinary Bayesian Mode (Omega + Omicron + Core synthesis)

Lifecycle assistants live here:

LogiCore → operations lifecycle assistants (job/quote/invoice lifecycle)

CreatiCore → marketing lifecycle assistants (campaign lifecycle)

Structured outputs may include:

JobPlan

VisitPlan

TaskList

MessageDraft

InvoiceActionPlan

CampaignPlan

Rule:

Titan Zero never writes directly to tz_ tables.

It proposes actions; execution happens via the core engines (Work/Money/Talk/etc) and is mediated by governance.

Core components:

engine_router

model_drivers

assistant_profiles

artifact_generators

prompt_registry

11. Titan Analytics Engine

The business intelligence engine.

Responsible for:

dashboards

KPIs

operational metrics

financial reporting

forecasting and performance

Core tables (tz_):

tz_metrics

tz_metric_snapshots

tz_analytics_queries

Rule:

Analytics consumes signals as its primary input (not raw table scans).

Minimum Engine Set (MVP)

If you want the smallest working Titan platform, install these first:

1. Titan Identity Engine
2. Titan Customer Engine
3. Titan Work Engine
4. Titan Evidence Engine
5. Titan Money Engine
6. Titan Talk Engine
7. Titan Signal Engine

8. Titan Pulse Engine

9. Titan Zero Engine

That combination can run an entire service business end-to-end.

Real Titan Execution Model

User



Interfaces (Command / Talk / Origin / Go)



Assistants (LogiCore / CreatiCore)



Titan Zero Engine (governance + routing + approvals)



Core Engines (Work/Money/Talk/Evidence/Docs/Customer)



tz_ * Operational Tables

And Titan Pulse Engine listens to Titan Signal Engine for automation triggers across everything.